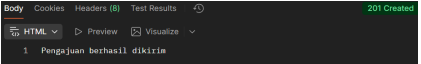
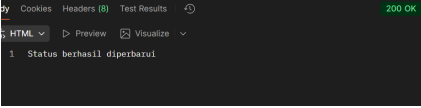
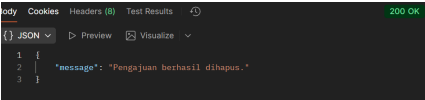


Method	Endpoint	Deskripsi	Request Body (JSON)	Response Sukses (200 OK)
POST	/api/login	Digunakan oleh admin untuk masuk ke dashboard.	{ "usernam e": "admin@"	

			dishub.bli tarkab.go .id", "passwor d": "passwor d_rahasia " }	
POST	/api/pengajuan	Mengirim data formulir dan upload berkas warga.	nama_lengkap: (Text) nik: (Text - 16 digit) alamat: (Text) nomor_w hatsapp: (Text) jumlah_peserta: (Int) file_ktp: (File) file_pernyataan: (File) file_permohonan: (File): file_berkas_izin: (File): file_denda_lokasi: (File):	
GET	/api/pengajuan	Melihat semua daftar pengajuan masuk.	Authoriza tion: Bearer <TOKEN _JWT>	
PUT	/api/pengajuan/:id/status	Mengubah status menjadi Disetujui atau Ditolak.	{ "status": "Disetujui" } }	

DELETE	/api/pengajuan/:id	Menghapus data dan file terkait secara permanen.	Authorization: Bearer <TOKEN_JWT>	
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3. Dokumentasi Kode Sumber (Source Code)

a. Struktur Folder

```
userdishub@mbaklina:~/var/www/mbaklina$ ls -l
total 24
drwxrwxr-x 4 userdishub userdishub 4096 Dec  1 13:32 backend
-rw-rw-r-- 1 userdishub userdishub 3374 Nov 24 10:34 db_penutupan_jalan.sql
-rw-rw-r-- 1 userdishub userdishub 1515 Dec  1 14:41 docker-compose.yml
-rw-rw-r-- 1 userdishub userdishub 1461 Dec  1 14:22 docker-compose.yml.save
drwxrwxr-x 6 userdishub userdishub 4096 Dec  1 13:34 frontend
drwxr-xr-x 6 dnsmasq      root          4096 Dec  1 14:25 mysql_data
```

b. Snippet Backend

```
// --- Endpoint Publik ---
const storage = multer.diskStorage({
  destination: (req, file, cb) => cb(null, 'uploads/'),
  filename: (req, file, cb) => cb(null, `${req.body.nik}-${Date.now()}-${file.originalname}`)
});
const upload = multer({
  storage: storage,
  limits: {
    fileSize: 2 * 1024 * 1024 // 2 Megabytes
  }
});

app.post('/api/pengajuan', upload.fields([
  { name: 'file_ktp', maxCount: 1 }, { name: 'file_pernyataan', maxCount: 1 },
  { name: 'file_permohonan', maxCount: 1 }, { name: 'file_berkas_izin', maxCount: 1 },
  { name: 'file_denah_lokasi', maxCount: 1 }
]), (req, res) => {
```

```
// --- Endpoint Otentikasi ---
app.post('/api/login', (req, res) => {
  const { username, password } = req.body;
```

```
// API Endpoint untuk MENGHAPUS Pengajuan (untuk Admin)
app.delete('/api/pengajuan/:id', verifyToken, (req, res) => {
  const { id } = req.params;
```

c. Snippet Frontend

```
const handleSubmit = async (e) => {
  e.preventDefault();
  setError('');
  setIsLoading(true);

  if (formData.nik.length !== 16) {
    setError('NIK harus terdiri dari 16 digit angka.');
    setIsLoading(false);
    return;
  }

  const data = new FormData();
  Object.keys(formData).forEach(key => data.append(key, formData[key]));
  Object.keys(files).forEach(key => data.append(key, files[key]));
```

d. Konfigurasi Docker

```
# 4. ModSecurity WAF + Nginx
waf:
  image: owasp/modsecurity-crs:nginx
  restart: always
  ports:
    - "80:80" # Pintu Utama (Publik)
  environment:
    - PROXY=1
    - BACKEND=http://frontend:80
    # Jika url mengandung /api, lempar ke Backend
    - LOCATION_API=/api
    - PROXY_API=http://backend:5000
    # Mode Deteksi
    - MODSEC_RULE_ENGINE=On
  depends_on:
    - frontend
    - backend
  networks:
    - jaringan-mbaklina

networks:
  jaringan-mbaklina:
```

4. Struktur SQL Database

```
mysql> DESCRIBE pengajuan;
```

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
nama_lengkap	varchar(255)	NO		NULL	
nik	varchar(16)	NO		NULL	
alamat	text	NO		NULL	
nomor_whatsapp	varchar(20)	NO		NULL	
jumlah_peserta	int(11)	NO		NULL	
file_ktp	varchar(255)	NO		NULL	
file_pernyataan	varchar(255)	NO		NULL	
file_permohonan	varchar(255)	NO		NULL	
file_berkas_izin	varchar(255)	NO		NULL	
file_denah_lokasi	varchar(255)	NO		NULL	
status	enum('Menunggu', 'Disetujui', 'Ditolak')	YES		Menunggu	
tanggal_pengajuan	timestamp	NO		CURRENT_TIMESTAMP	

13 rows in set (0.00 sec)

```
mysql> DESCRIBE admins;
```

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
username	varchar(100)	NO	UNI	NULL	
password_hash	varchar(255)	NO		NULL	
created_at	timestamp	NO		CURRENT_TIMESTAMP	

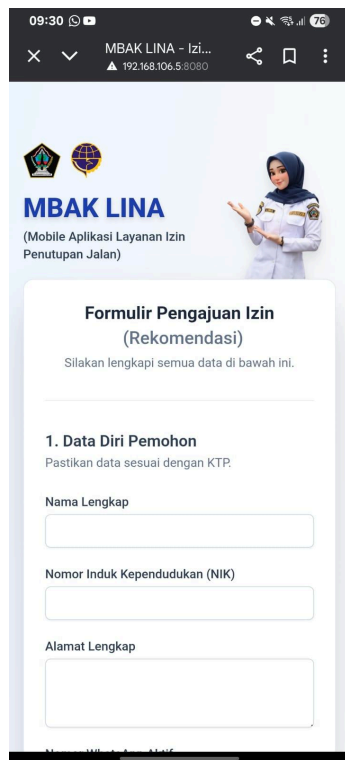
4 rows in set (0.00 sec)

5. Dokumen User Manual

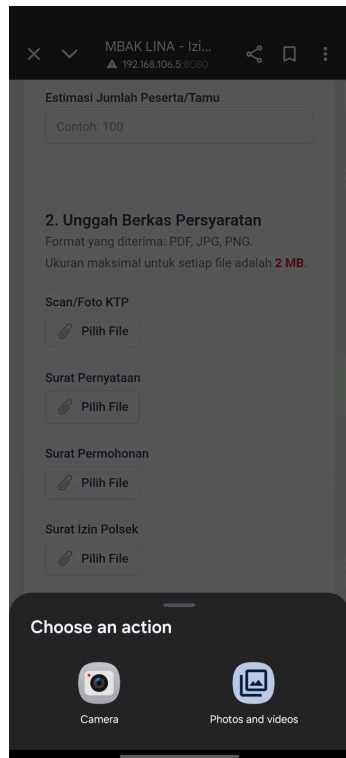
- Halaman Utama (Desktop):



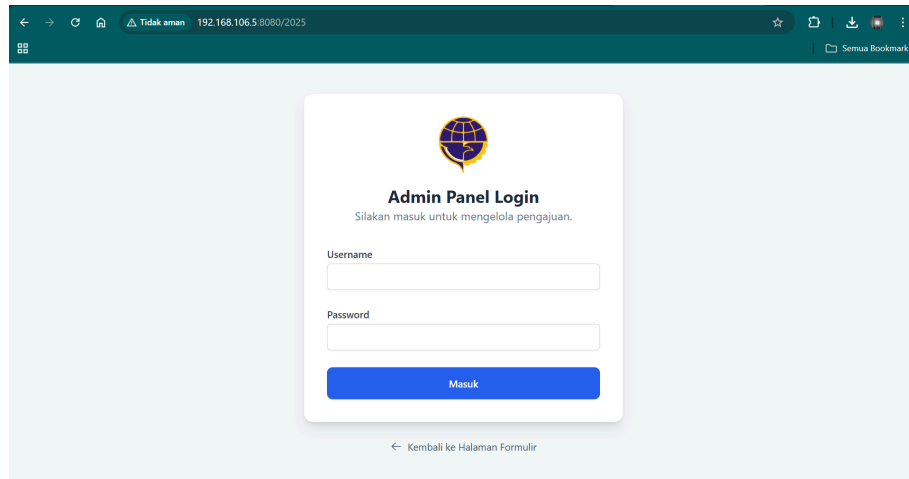
b. Halaman Utama (Mobile):



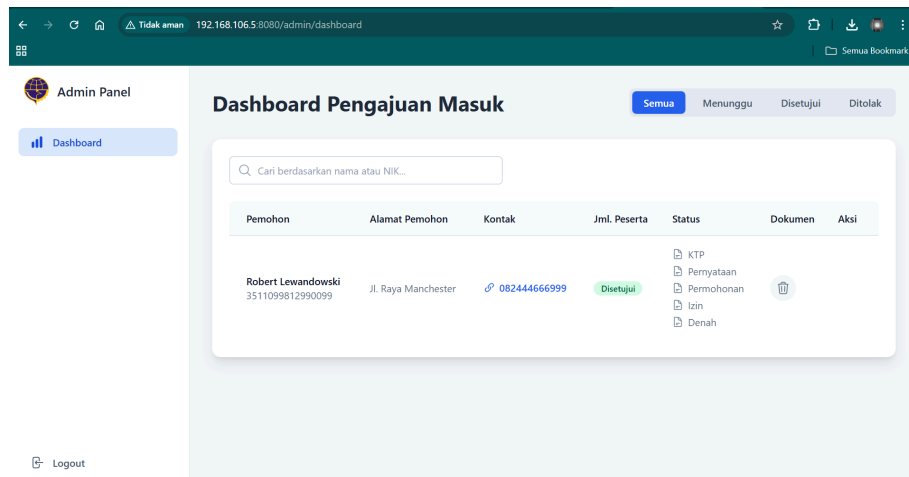
c. Fitur Upload:



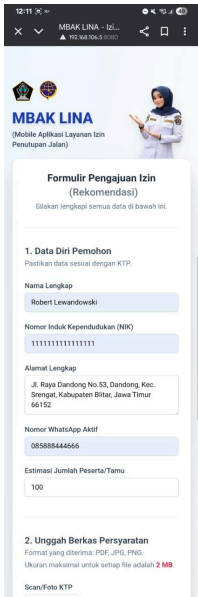
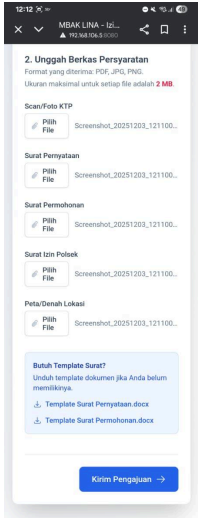
d. Login Admin:



e. Dashboard Admin:

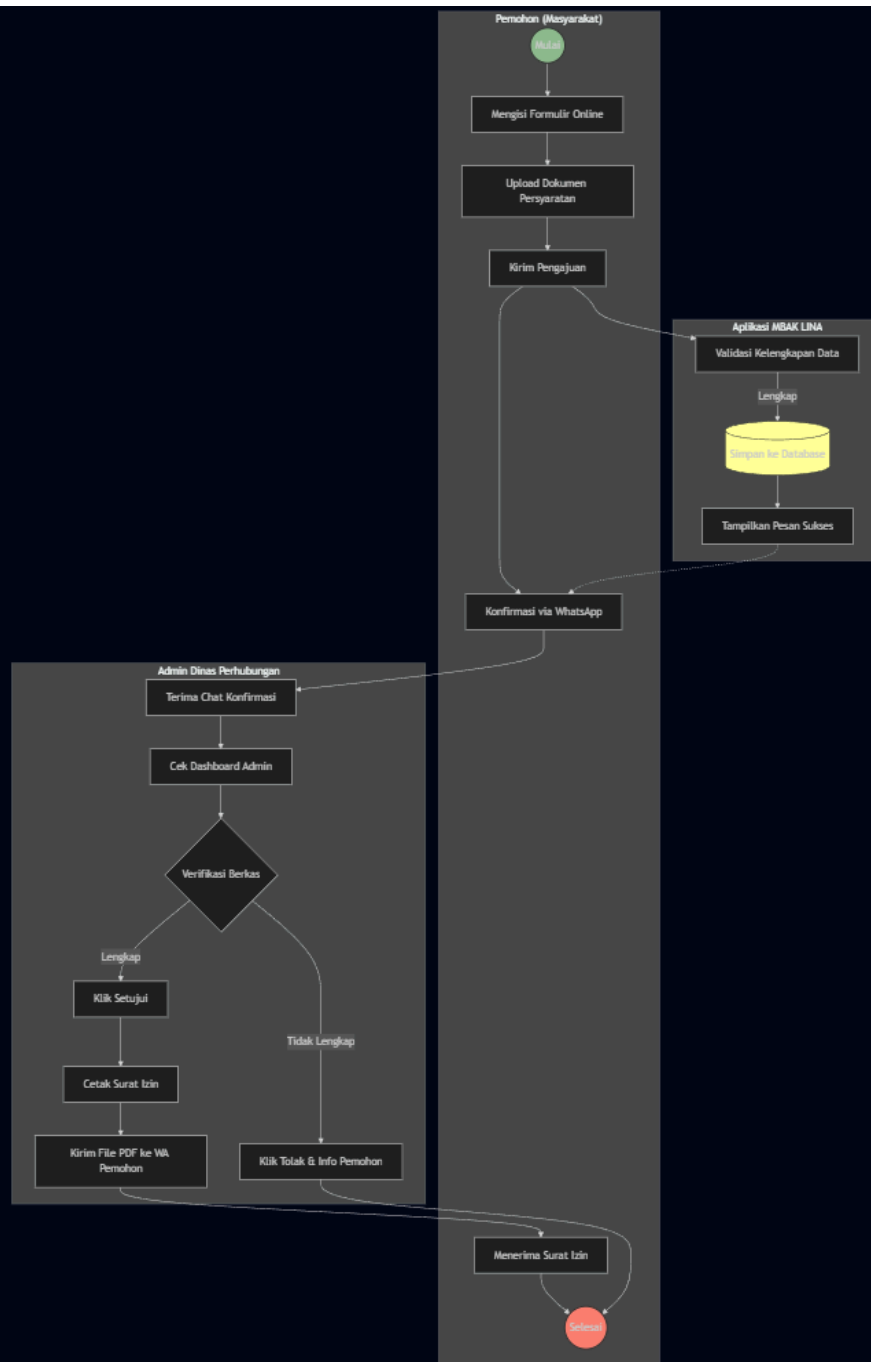


6. Dokumen Hasil Pengetesan (Functional Test)

No	Fitur	Skenario Tes	Hasil (Pass/Fail)	Bukti Gambar
1	Input Data	Mengisi semua kolom wajib	Pass	
2	Upload File	Upload JPG & PDF	Pass	

3	Keamanan	Akses port 5000 langsung dari browser	Pass (blocked)	 The screenshot shows a mobile browser interface with a dark theme. At the top, the status bar displays the time 12:18 and various icons. Below the status bar, there is a navigation bar with a back arrow, a search icon, and a menu icon. The main content area displays an error message: 'Error: GET /' and 'Cannot GET /'. The error message is in white text on a dark background. At the bottom of the screen, there is a home indicator bar.
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7. Dokumen Business Process (Notasi BPMN)



8. Dokumen Metadata Arsitektur Aplikasi

- a. Tabel admins: Menyimpan kredensial petugas (password terenkripsi Bcrypt).

```
mysql> DESCRIBE admins;
```

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
username	varchar(100)	NO	UNI	NULL	
password_hash	varchar(255)	NO		NULL	
created_at	timestamp	NO		CURRENT_TIMESTAMP	

4 rows in set (0.00 sec)

- b. Tabel pengajuan: Menyimpan data transaksi permohonan

```
mysql> DESCRIBE pengajuan;
```

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
nama_lengkap	varchar(255)	NO		NULL	
nik	varchar(16)	NO		NULL	
alamat	text	NO		NULL	
nomor_whatsapp	varchar(20)	NO		NULL	
jumlah_peserta	int(11)	NO		NULL	
file_ktp	varchar(255)	NO		NULL	
file_pernyataan	varchar(255)	NO		NULL	
file_permohonan	varchar(255)	NO		NULL	
file_berkas_izin	varchar(255)	NO		NULL	
file_denah_lokasi	varchar(255)	YES		NULL	
status	enum('Menunggu', 'Disetujui', 'Ditolak')	YES		Menunggu	
tanggal_pengajuan	timestamp	NO		CURRENT_TIMESTAMP	

13 rows in set (0.00 sec)

9. Dokumen BCP (Business Process Continuity Plan)

No	Skenario Gangguan	Dampak	Prosedur Pemulihan (Recovery Plan)
1	Layanan Terhenti (Container Crash) Aplikasi tidak bisa diakses, muncul error 502/504.	Warga tidak bisa akses web. Admin tidak bisa login.	1. Restart Container: Masuk terminal server dan jalankan: sudo docker-compose restart 2. Cek Status: Pastikan semua status "Up" dengan: sudo docker ps
	Server Restart Mendadak VPS mati karena listrik atau maintenance provider.	Semua layanan mati total.	Otomatis Pulih: Docker telah dikonfigurasi dengan restart: always . Layanan akan menyala otomatis begitu server hidup kembali. Admin hanya perlu memantau akses web.
	Data Hilang / Database Korup Kesalahan hapus	Data pengajuan hilang.	Restore Database: Mengembalikan data dari file backup terakhir (backup.sql) ke dalam

	data atau disk error.		container database. Command: <i>cat backup.sql</i>
	Bencana Total (Server Hancur/Terhapus)	Seluruh sistem hilang.	Re-Deployment: 1. Sewa VPS baru. 2. Install Docker. 3. Pull Source Code dari Gitlink. 4. Jalankan <i>docker-compose up -d.</i> 5. Restore database dari backup terakhir.

a. Simulasi Backup dan Cek File Backup:

```

userdishub@mbaklina:~$ sudo docker exec mbaklina_db_1 mysqldump -u admin_mbaklina -pDishub123 --no-tablespaces db_penutupan_jalan > backup_harian.sql
mysqldump: [Warning] Using a password on the command line interface can be insecure.
userdishub@mbaklina:~$ ls -lh backup_harian.sql
-rw-rw-r-- 1 userdishub userdishub 3.7K Dec  3 13:58 backup_harian.sql

```